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PI Name:	Rithidech, Kanokporn Ph.D.		
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PI Email:	Kanokporn.Rithidech@sbumed.org	**Fax:**	FY (631) 444-3424
PI Organization Type:	UNIVERSITY	**Phone:**	(631) 444-3446
Organization Name:	State University New York at Stony Brook		
PI Address 1:	Department of Pathology		
PI Address 2:	BHS T9 Health Sciences Center		
PI Web Page:			
City:	Stony Brook	**State:**	NY
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Task Description:

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Different types of galactic cosmic rays (GCR) are present in space and have large mass and energy (HZE particles). Among these, stripped 56Fe ions are of major concern. Although HZE particles are approximately 1% of GCR, their contribution to health risk could be significant because of their very high linear energy transfer (LET) resulting in a larger amount of energy being deposited in the hit cells. To protect astronauts in space environments, it is essential to improve our knowledge of radiation-induced damage associated with health risk of late occurring diseases (such as cancer).

It has been well established that acute myeloid leukemia is one of the major malignancies associated with exposure to radiation in human beings and mice. It is therefore considered as one of the most important cancers related to space flights. For radiation-induced genetic damage to be useful for risk assessment, it must be established in in vivo systems especially in cells that are known to be at risk for health problems associated with radiation exposure such as bone marrow cells, the known target cell for radiation-induced leukemia.

We exposed male CBA/CaJ mice whole body to varying doses of 1 GeV/amu 56Fe ions or 137Cs gamma rays. We collected bone marrow (BM) cells from exposed mice at different times up to 3 months post-exposure. The mouse in vivo micronucleus (MN) assay was used to evaluate chromosomal damage that was induced in the bone marrow but the actual analysis was done in blood erythrocytes. Moreover, the novel technology for genome-wide multicolor fluorescence in situ hybridization of mouse chromosomes (mFISH) was used for the detection, analysis, and characterization of chromosomal damage in metaphase cells prepared from the BM.

Task Objective: Our in vivo study was aimed to address the following hypotheses: (1) Complex chromosomal exchanges are more prevalent in the BM cells of mice exposed to 56Fe ions than those exposed to similar doses of 137Cs gamma rays. (2) There is a dose-dependent increase in the frequency of chromosome aberrations in BM cells of mice exposed to 56Fe ions or 137Cs gamma rays. (3) Genomic instability (expressed as delayed appearance of chromosome aberrations) is detected in a fraction of surviving cells at later times post-irradiation.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Our data will definitely provide pivotal information that should lead to a better understanding of dose-response kinetics and of mechanisms by which 56Fe ions induced chromosome aberrations that can be potentially involved in the induction of genomic instability and cancer. This statement is based upon the certainty that chromosome aberrations have been proven to be the most relevant endpoint for detecting carcinogenic activity of suspect agents because of their strong correlation with the induction of cancers. The results also would be important in improving risk assessment of protracted space flight. Both 137Cs gamma rays (the reference radiation) and 56Fe ions are used in this study. The results would therefore facilitate determining whether radiation quality influences the probabilities for increased carcinogenesis from space radiation. Moreover, chromosomal damage is measured at the low-dose level of 56Fe ions (i.e. 0.1Gy). Consequently, the results should allow us to make a meaningful comparison for understanding the in vivo RBE of 56Fe ions on the target cells as a function of absorbed dose, aberration type and time. The results also would lead to a better understanding of basic mechanisms of cytogenetic damage potentially involved in cancer induction by these two different sources of radiation.

Task Progress
A. Analysis of Micronucleus in Mouse Blood Erythrocytes: Effects of 56Fe ions and 137Cs gamma rays on the frequency of MN-PCEs and MN-NCEs at day 7 post-irradiation.

In mice exposed to 56Fe ions, a slight increase in the frequencies of MN-PCEs was detected at day 7 in those exposed to high doses (>0.1 Gy) of 56Fe ions. In contrast, there were no increases in MN-PCEs frequencies in blood samples collected at day 7 post-irradiation in mice exposed to any dose-level of 137Cs gamma rays, as compared to that in sham controls. Moreover, the frequencies of MN-PCEs in mice exposed to 137Cs gamma rays were lower than that in their corresponding controls, regardless of radiation dose. Such a decrease was more pronounced in mice exposed to the highest dose of 137Cs gamma rays. We detected a dose-dependent increase in the frequencies of MN-NCEs at day 7 after exposure to either type of radiation. The frequencies of MN-NCEs induced by 56Fe ions were slightly higher than those induced by 137Cs gamma rays. We used both linear and linear-quadratic regression equations to test the fitness of the MN data. However, the use of a linear–quadratic regression equation did not improve the fit of the data, so the simpler linear equation was used to describe the dose-response curves. The RBE value for 56Fe ions (related to that induced by 137Cs gamma rays) for in vivo induction of MN-NCEs at day 7 was 3.4.

Effects of 56Fe ions and 137Cs gamma rays on the frequency of MN-PCEs and MN-NCEs at 3 months post-irradiation.

In contrast to samples collected at day 7, we observed dose-dependent increases in levels of micronuclei in the immature PCEs in blood cells collected at 3 months following whole body exposure of CBA/CaJ mice to either 56Fe ions or 137Cs gamma rays. This finding indicates the potential in vivo induction of genomic instability in CBA/CaJ by both 56Fe ions and 137Cs gamma rays. This finding indicates the potential in vivo induction of genomic instability in CBA/CaJ mice by both 56Fe ions and 137Cs gamma rays. A linear equation was used to describe the dose-response curves. The RBE value for the in vivo induction of MN-PCEs for 56Fe ions (related to that induced by 137Cs gamma rays) was 2.6 at 3 months post-irradiation. With respect to MN-NCEs, there were no differences in MN-NCEs levels in exposed mice related to the control levels at 3 months post-exposure, regardless of radiation type. It is not surprising that there were no increases in MN-NCEs at 3 months because of the limited life span of micronucleated NCE of the mouse, i.e. approximately 30 days.

In summary, our mouse in vivo blood micronucleus data demonstrated that:

(1) At day 7, there was a delay in expression of damage after exposure of CBA/CaJ mice to 56Fe ions.

(2) 56Fe ions and gamma rays induced in vivo expression of genomic instability in exposed CBA/CaJ mice.

(3) A single dose of 1 Gy 56Fe ions or 3 Gy 137Cs gamma rays induced a similar level of chromosomal damage and genomic instability in hematopoietic cells of exposed CBA/Ca mice.

(4) RBE values changed with time following irradiation.

Task Progress:	B. Analysis of Metaphase Chromosome Aberrations in Mouse Bone Marrow Cells by Means of Multicolor Fluorescence in situ Hybridization (mouse mFISH): Our study is the first to use the mouse, whole-genome, multi-color FISH (mFISH) technique to evaluate cytogenetic effects of radiation in bone marrow cells after a whole-body exposure of mice to 56Fe ions or 137Cs gamma rays. Early Time Point: We have completed the analysis of chromosome aberrations in metaphase cells prepared from the bone marrow collected at day 7 after exposure of CBA/CaJ mice to graded doses of 56Fe ions (0, 0.1, 0.5, and 1.0 Gy) or 137Cs gamma rays (0. 0.5, 1.0, and 3.0 Gy). The results have been published [Rithidech et al, Radiat Environ Biophys, 46(2):137-145, 2007]. Currently, a revised version of the manuscript has been submitted to the journal. At day 7 post exposure, several types of aberrations were observed in bone marrow cells collected from mice exposed to either 56Fe ions or 137Cs gamma rays. These were exchanges and breaks (both chromatid- and chromosome- types). Chromosomal exchanges included translocations (Robertsonian or centric fusion, reciprocal and incomplete types), insertions, and dicentrics. Most of the translocations belong to the one-way type. In general, we found no evidence of a non-random involvement of specific chromosomes in any type of aberration observed in mice exposed to 56Fe ions or 137Cs gamma rays. We observed, however, abnormalities (chromatid-breaks and translocations) involving chromosome 2 in metaphase cells prepared from BM cells collected from some mice exposed to 3.0 Gy of 137Cs gamma rays or 1.0 Gy of 56Fe ions only. Additionally, high frequencies of metaphase cells with an abnormal chromosome 14 (similar types of damage as detected on chromosome 2) were detected in mice exposed to these two high doses of radiation (6-fold higher than in mice exposed to smaller doses of either type of radiation). The highest dose of 56Fe ions (1.0 Gy) induced more cells with multiple aberrations, as compared to those induced by the high dose of 137Cs gamma rays (3.0 Gy). One of the CBA/CaJ mice exposed to 3 Gy of 137Cs gamma rays acquired two cells with translocations involving chromosomes 2 and 3 [t(2;3)] suggesting a clonal expansion of this specific translocation. Such clonal expansion was not detected in 56Fe-ion-exposed mice. Furthermore, a cell containing t(2;3) was observed in another 137Cs gamma-ray-exposed mouse. At the radiation dose range used in our in vivo study, the majority of exchanges were simple. Complex exchanges were detected in BM cells collected from mice exposed to 1 Gy of 56Fe ions or 3 Gy of 137Cs gamma rays only, but their frequencies were low. Overall, our in vivo data indicate that the frequency of complex chromosome exchanges was not significantly different between bone marrow cells collected from mice exposed to 56Fe ions or 137Cs gamma rays. Each type of radiation induced statistically significant dose-dependent increases (ANOVA, $p<0.01$) in the frequencies of chromosomal damage, including the numbers of abnormal cells. We used both linear and linear-quadratic regression equations to test the fitness of the data, i.e. the number of abnormal cells, the number of breaks (both chromatid- and chromosome- types), and the number of exchanges (all types). In general, the linear-quadratic equations are a better fit with our data. Based upon the linear-terms of the linear-quadratic equations for dose-response curves, 56Fe ions were 1.6 (all types of exchanges), 4.3 (abnormal cells), and 4.2 (breaks) times more effective than 137Cs gamma rays in inducing chromosomal damage. Late Time Point: We also have finished the analysis of metaphase chromosomes prepared from bone marrow cells collected at 3 months after exposure of CBA/CaJ mice to 56Fe ions (0 or 1.0 Gy) or 137Cs gamma rays (0 and 3.0 Gy). For each radiation type, there were four mice per dose. Similar to samples collected at day 7 post-irradiation, several types of aberrations in bone marrow cells collected from mice exposed to either type of radiation were found. These were exchanges and breaks (both chromatid- and chromosome- types). Chromosomal exchanges included translocations (Robertsonian or centric fusion, reciprocal and one-way incomplete one types), and dicentrics. All exchanges were simple types. The frequencies of each type of chromosome aberrations were much lower that those found in cells collected at day 7 post-irradiation. Our data indicated significant increases in numbers of abnormal cells, breaks (both chromatid- and chromosome-types), and exchanges (all types) in mice exposed to 1.0 Gy 56Fe ions or 3.0 Gy 137Cs gamma rays related to those in their corresponding sham controls (ANOVA, $p<0.05$). These findings indicated that 1.0 Gy 56Fe ions or 3.0 Gy 137Cs gamma rays are capable of inducing genomic instability in bone marrow cells of exposed mice. However, there was no statistical difference between the frequencies of abnormal cells, breaks (both chromatid- and chromosome-types), and exchanges (all types) in mice exposed to 1.0 Gy 56Fe ions or 3.0 Gy 137Cs gamma rays. This finding indicates that the two doses of these two different types of radiation are equitoxic in inducing chromosome aberrations or genomic instability in bone marrow cells of exposed CBA/CaJ mice. In summary, we have compared in vivo cytogenetic effects of 56Fe ions to 137Cs gamma rays, using the mFISH method, in cells that are known to be at risk for adverse health effects. The data indicate that 56Fe ions are more effective in inducing chromosome damage (per unit dose of radiation) than 137Cs gamma rays (Rithidech et al, Radiat Environ Biophys, 46(2):137-145, 2007; Rithidech et al, In Press in Adv Space Res, 2007). The obtained results have important implications for risk assessment.
Bibliography Type:	Description: (Last Updated: 01/20/2026)
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